

Microeconomics I

Class 3 — Utility representation

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slide 1 of 28

Slide 1 Title

Class 3 covers one section of the textbook, MWG 3.C: when a number $u(x)$ can stand in for the preference relation $\succsim$, what that number does and does not mean, and why one familiar preference from class 2 has no such number at all.

Short quiz

8 minutes · pens down at 10:08

Circle the letter of your answer in the block at the top of the sheet.

Changed your mind? Cross out that circle and circle the new letter.

Write your name on the block, tear the strip off along the dashed line, hand in *only the strip*.

quiz page, not numbered

Quiz Start of class, on paper

A short quiz on paper opened the class: circle the letters of your answers in the block on the sheet, write your name on it, and hand in only the strip.

Plan for today

10:00	Short quiz
10:08	Practicalities; five stars or a thumb
10:15	Utility functions: what the numbers mean
10:28	Continuity, existence, the lexicographic case
10:50	Break
11:00	Quasiconcavity and concavity
11:23	Common utility functions
11:40	Summary; problem set 1

Slide 2 Plan for today

The morning at a glance. A real rating screen opens the class; then the definition of a utility function and the sense in which its numbers are only labels. Next: continuity, the theorem that guarantees a utility function, and the preference that has none. After the break: quasiconcavity and concavity, and the utility functions used for the rest of the month.

Everything today is MWG section 3.C; Muñoz-Garcia chapter 1 is the gentler read.

Practicalities

- Problem set 1 today, on paper: one copy per group, your group's own numbers
- Due in class Wed 14 Oct, one paper copy per group, all names on it
- Office hours: Wednesdays 12:00–13:00, after class
up to 20 minutes per student; those who book by e-mail have priority
- Slides and reading: pkocourek.com/teaching/micro-1 · pkocourek@ub.edu

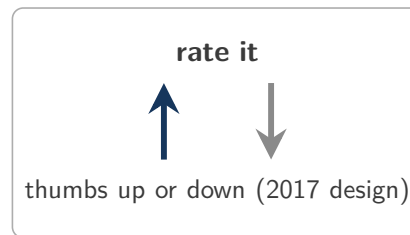
Slide 3 Practicalities

Problem set 1 is handed out on paper at the end of class, one copy per group. Every group has its own numbers, and the sheet says which results to box.

It is due in class on Wednesday 14 October: one paper copy per group, all names on the first page.

Office hours: after class on Wednesdays, up to 20 minutes per student; booking by e-mail gives priority. Slides and reading: pkocourek.com/teaching/micro-1 · pkocourek@ub.edu.

Five stars or a thumb?



Which of the two is a utility function?

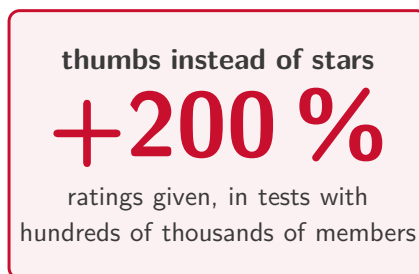
Netflix: stars until April 2017, thumbs since (a double thumb-up added 2022). Evidence, not a recommendation.

Slide 4 Five stars or a thumb?

Two ways a streaming service can ask you to rate a title: one to five stars, or a thumb up or down. The question to hold: which of the two is a utility function? It is answered on slide 10, once the definition is on the table.

The service is Netflix: stars until April 2017, thumbs since. This is evidence about how people answer a rating screen, not a recommendation of anything.

Thumbs instead of stars: three times as many answers



Stars invite **arithmetic**; a thumb does not. Does the theory need it? Back to it on slide 10.

Netflix, "Goodbye Stars, Hello Thumbs", 5 April 2017; announced 16 March 2017 (Variety, TechCrunch).

Slide 5 Thumbs instead of stars: three times as many answers

In 2016 Netflix tested thumbs against stars with hundreds of thousands of members: thumbs produced 200 % more ratings, three times as many. The change was announced in March 2017; the stars went in April 2017.

The star shown next to a title was never an average of other members' ratings. It was Netflix's prediction of what you would give it, personal to you. Members read it as an average anyway, so it was replaced by a personal "% match", also a prediction.

Stars invite arithmetic: averaging, comparing gaps. A thumb does not. The question for the theory, answered on slide 10: does it need that arithmetic? Caveat: the 200 % says people answered more readily; it says nothing about the structure of anyone's preferences.

Definition: a utility function represents $\succsim$

$u : X \rightarrow \mathbb{R}$ **represents** $\succsim$ if, for all $x, y \in X$,

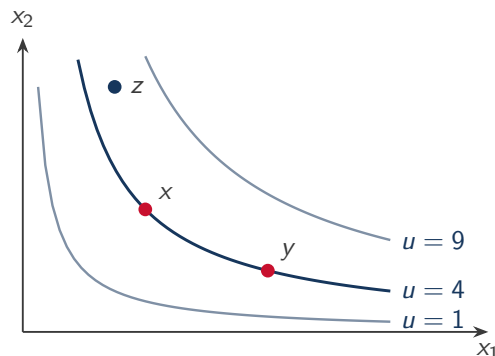
$$x \succsim y \iff u(x) \geq u(y)$$

Represented $\Rightarrow \succsim$ is rational

complete and transitive, because $\geq$ on $\mathbb{R}$ is

MWG Definition 1.B.2, Proposition 1.B.2.

$$u(x) = x_1 x_2: \quad u(x) = u(y) = 4, \text{ so } x \sim y; \quad u(z) = 6, \text{ so } z \succ x.$$



Slide 6 Definition: a utility function represents $\succsim$

MWG Definition 1.B.2. A function $u : X \rightarrow \mathbb{R}$ represents $\succsim$ if, for every pair of bundles, x is at least as good as y exactly when $u(x) \geq u(y)$: the number ranks bundles in the same order as the preference does.

Picture: $u(x) = x_1 x_2$, with the curves $x_1 x_2 = 1, 4, 9$. The bundles $x = (2, 2)$ and $y = (4, 1)$ both score 4, so $x \sim y$; $z = (1.5, 4)$ scores 6, so $z \succ x$.

A represented preference is automatically rational (Proposition 1.B.2): $\geq$ on the real numbers is complete and transitive, so $\succsim$ inherits both. The notation $u : \mathbb{R}_+^2 \rightarrow \mathbb{R}$ from class 1 now has its meaning.

Slide 7 From class 2: four of the five were already “ranked by a number”

	$x \succsim y \iff$	$u(x) =$
perfect substitutes	$x_1 + x_2 \geq y_1 + y_2$	$x_1 + x_2$
perfect complements	$\min\{x_1, x_2\} \geq \min\{y_1, y_2\}$	$\min\{x_1, x_2\}$
Cobb–Douglas type	$x_1 x_2 \geq y_1 y_2$	$x_1 x_2$
quasilinear	$x_1 + \sqrt{x_2} \geq y_1 + \sqrt{y_2}$	$x_1 + \sqrt{x_2}$
lexicographic	$x_1 > y_1$, or $x_1 = y_1$ and $x_2 \geq y_2$	?

Three questions for today: is the number **unique**? When does one **exist**? What do its **properties** say about $\succsim$?

Four of the five preferences of class 2 were written as “ $x \succsim y$ if and only if some number computed from x is at least the same number computed from y ”. That number is a utility function: $x_1 + x_2$, $\min\{x_1, x_2\}$, $x_1 x_2$, $x_1 + \sqrt{x_2}$.

The lexicographic preference had single points as indifference sets. Whether it has a number is the middle of today (from slide 14 on).

Three questions, answered in this order: is the number unique (slides 8–9)? When does one exist (slides 12–17)? What do its properties say about $\succsim$ (slides 21–24)?

In pairs: same preference as $u(x) = x_1x_2$?

$v(x) =$	same $\succsim$?	
$\sqrt{x_1x_2}$	✓	$v = f(u)$ with $f(t) = \sqrt{t}$, strictly increasing
$2 - \frac{1}{1+x_1x_2}$	✓	$f(t) = 2 - \frac{1}{1+t}$, strictly increasing; all values below 2
$x_1 + x_2$	✗	(1, 4) and (2, 2): u says $\sim$, v says $5 > 4$
$x_1x_2^2$	✗	(1, 4) and (2, 2): u says $\sim$, v says $16 > 8$

Four minutes. Same preference means: the same ranking of every pair of bundles.

Two keep the ranking, one of them bounded. Two disagree already at (1, 4) vs (2, 2).

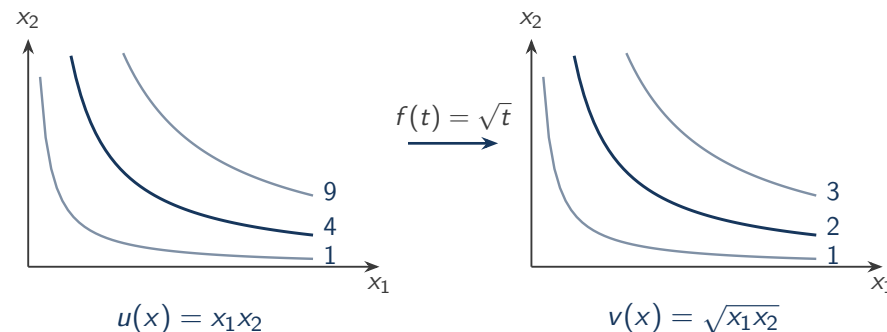
Slide 8 In pairs: same preference as $u(x) = x_1x_2$?

Try it: does v rank every pair of bundles the way $u(x) = x_1x_2$ does? For a “yes”, find a strictly increasing f with $v = f(u)$; for a “no”, two bundles the two functions rank differently.

$\sqrt{x_1x_2}$: yes, $f(t) = \sqrt{t}$. $2 - 1/(1+x_1x_2)$: yes, $f(t) = 2 - 1/(1+t)$ is strictly increasing. Its values all lie below 2, so a utility function for this preference can be bounded.

$x_1 + x_2$: no. Take (1, 4) and (2, 2): u gives 4 and 4, indifferent, but v gives $5 > 4$. $x_1x_2^2$: no, the same pair gives $16 > 8$. Curves of a similar shape are not enough; its curves have slope $-x_2/(2x_1)$ instead of $-x_2/x_1$, a different map.

The numbers are labels: any strictly increasing relabelling will do



u represents $\succsim$ and $f : \mathbb{R} \rightarrow \mathbb{R}$ strictly increasing $\Rightarrow v = f \circ u$ represents $\succsim$

Utility is **ordinal**: same map, different numbers on the curves. Differences of u mean nothing.

MWG 3.C, after Proposition 3.C.1; Section 1.B. Proof: $v(x) \geq v(y) \iff u(x) \geq u(y) \iff x \succsim y$.

Slide 9 The numbers are labels: any strictly increasing relabelling will do

If u represents $\succsim$ and f is strictly increasing, then $v = f \circ u$ represents $\succsim$ too: $v(x) \geq v(y)$ exactly when $u(x) \geq u(y)$, exactly when $x \succsim y$. The picture shows the same three curves labelled 1, 4, 9 by u and 1, 2, 3 by $v = \sqrt{u}$.

Utility is ordinal: the numbers are labels for the ranking. Differences, ratios and sums of utilities carry no information, because another equally valid relabelling changes them all.

Why must f be strictly increasing? With $f(t) = \min\{t, 4\}$, every bundle with $x_1x_2 \geq 4$ gets the same label 4, so v no longer separates bundles that $\succsim$ ranks differently: it does not represent $\succsim$.

Back to the stars

stars invite arithmetic

★★★★★ + ★★★★★ = 3?

averages and gaps: a **cardinal** use

a thumb

liked it, or not

nothing to add up: **ordinal** only

The theory needs only the ranking. Nothing in $\succsim$ says how much better.

Neither screen is a utility function: both coarsen the ranking, like $f(t) = \min\{t, 4\}$. Netflix's members read a personal prediction as an average: arithmetic the theory never does.

Slide 10 Back to the stars

Stars invite arithmetic: averaging ratings, treating the gap from 4 to 5 like the gap from 1 to 2, adding different people's numbers. That is a cardinal use, the one review sites make. A thumb says liked it or not, a label with nothing to add up: ordinal.

Answer to slide 4: neither screen is a utility function. Both coarsen the ranking, like $f(t) = \min\{t, 4\}$ on slide 9: titles you rank differently get the same score.

Netflix's star was a personal prediction, and members read it as an average: a cardinal reading of a number that only carried a ranking. The theory needs only the ranking; nothing in $\succsim$ says how much better.

Slide 11 Continuity and existence

Part 2: when does a utility function exist? Continuity, the theorem, and the preference that breaks it. MWG 3.C.

Continuity and existence

MWG 3.C

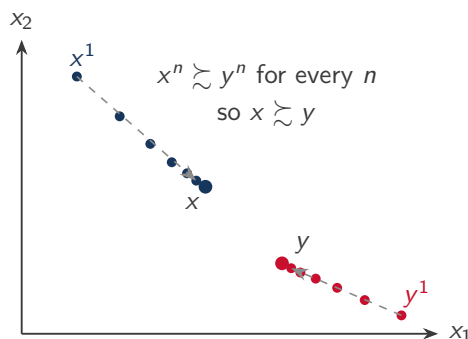
Definition: continuous preferences

$\succsim$ on X is **continuous** if it is preserved under limits:

$$\begin{aligned} & x^n \succsim y^n \text{ for all } n, \\ & x = \lim x^n, \quad y = \lim y^n \\ \Rightarrow & x \succsim y \end{aligned}$$

MWG Definition 3.C.1.

No sudden reversal at the limit: what holds along the whole sequence holds at the end.

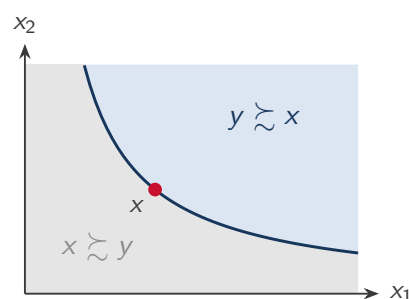


Slide 12 Definition: continuous preferences

MWG Definition 3.C.1. $\succsim$ is continuous if it is preserved under limits: whenever $x^n \succsim y^n$ for every n and the two sequences converge, $x = \lim x^n$ and $y = \lim y^n$, then $x \succsim y$.

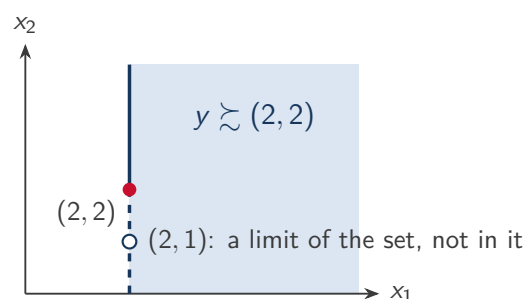
Picture: two convergent sequences, every pair ranked the same way, so the limits are ranked the same way too. In words: no sudden reversal at the limit; what holds along the whole sequence holds at the end.

Equivalently: upper and lower contour sets are closed



continuous

the curve belongs to both sets: closed



lexicographic

the dashed edge is missing: not closed

Closed: limits of points of the set stay in the set.

Right: $(2 + \frac{1}{n}, 1) \succsim (2, 2)$ for every n , but the limit $(2, 1)$ is not in the set.

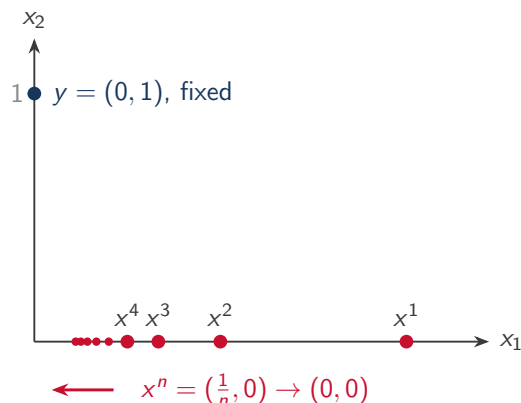
MWG 3.C, text after Definition 3.C.1 (one direction; the converse is Exercise 3.C.3).

Slide 13 Equivalently: upper and lower contour sets are closed

Equivalent: for every x , the upper contour set $\{y : y \succsim x\}$ and the lower contour set $\{y : x \succsim y\}$ are closed, that is, limits of points in the set stay in the set. One direction is quick: take $x^n = x$ fixed and $y^n \rightarrow y$ with $x \succsim y^n$; continuity gives $x \succsim y$, so the lower contour set is closed, and the upper one likewise. The converse is MWG Exercise 3.C.3.

Left: a continuous preference; the curve belongs to both sets. Right: the lexicographic upper contour set of $(2, 2)$ from class 2: everything with $y_1 > 2$, plus the half-line straight above $(2, 2)$. The points $(2 + \frac{1}{n}, 1)$ are in it and converge to $(2, 1)$, which is not: a tie in good 1, then $1 < 2$. The dashed edge is missing, so the set is not closed.

Lexicographic preferences reverse at the limit



$x^n \succ y$ for every n

$\frac{1}{n} > 0$: more of good 1 decides

$\lim x^n = (0, 0) \prec (0, 1) = y$

tie in good 1: now good 2 decides

Not continuous.

The ranking flips exactly where the first coordinates become equal. Example 3.C.1, continued.

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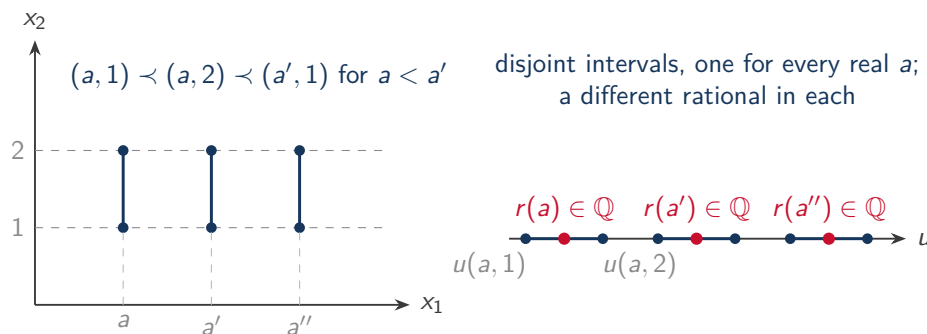
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Slide 14 Lexicographic preferences reverse at the limit

MWG Example 3.C.1. Fix $y = (0, 1)$ and let $x^n = (\frac{1}{n}, 0)$. For every n , $x^n \succ y$: good 1 decides, and $\frac{1}{n} > 0$. But $x^n \rightarrow (0, 0)$, and $(0, 0) \prec (0, 1)$: now the first coordinates tie and good 2 decides.

A preference reversal at the limit, exactly where the first coordinates become equal: the lexicographic preference is not continuous. So the existence theorem of slide 16 does not apply to it; whether some utility function exists anyway is the next slide.

Lexicographic preferences have no utility function at all



A one-to-one map from the reals into the rationals: uncountable into countable. Impossible.

MWG Example 3.C.1, small print on p. 46. This rules out every u , continuous or not.

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Slide 15 Lexicographic preferences have no utility function at all

A stronger statement, MWG p. 46: no function at all represents the lexicographic preference, continuous or not.

Suppose u did. For every real a , $(a, 1) \prec (a, 2)$, so $u(a, 1) < u(a, 2)$: a proper interval on the u -axis; pick a rational number $r(a)$ inside it. For $a < a'$, $(a, 2) \prec (a', 1)$ because good 1 decides, so $u(a, 2) < u(a', 1)$: the intervals do not overlap, and $r(a) \neq r(a')$.

That is a one-to-one map from the reals into the rationals, from an uncountable set into a countable one: impossible. Note what this preference does have: it is complete, transitive, strongly monotone and strictly convex (class 2), and still has no utility function.

Continuity is enough: a continuous utility function exists

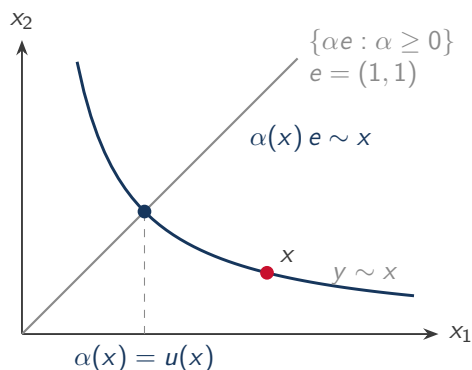
$\succsim$ rational and continuous on X
 $\Rightarrow$ some continuous $u : X \rightarrow \mathbb{R}$
 represents $\succsim$

Proof idea, $\succsim$ monotone:

$$u(x) = \alpha(x) \text{ with } \alpha(x)e \sim x$$

MWG Proposition 3.C.1, Figure 3.C.1 (monotone case).

Each curve is labelled by where it crosses the diagonal; further out crosses further along.



Slide 16 Continuity is enough: a continuous utility function exists

MWG Proposition 3.C.1: if $\succsim$ is rational and continuous, some continuous u represents it. The book proves the monotone case (Figure 3.C.1).

Proof idea: let $e = (1, 1)$ and look at the diagonal ray $\{\alpha e : \alpha \geq 0\}$. For each x , let $\alpha(x)$ be the scalar with $\alpha(x)e \sim x$, and set $u(x) = \alpha(x)$. In the picture the curve through $x = (4, 1)$ crosses the diagonal at $(2, 2)$, so $u(x) = 2$. Each curve is labelled by where it crosses the diagonal; monotonicity puts the labels in the right order.

Continuity only matters with infinitely many bundles: on a finite X , any rational $\succsim$ is represented by $u(x)$ = the number of bundles in the lower contour set of x (MWG Exercise 1.B.5).

Slide 17 Why every curve crosses the diagonal, once

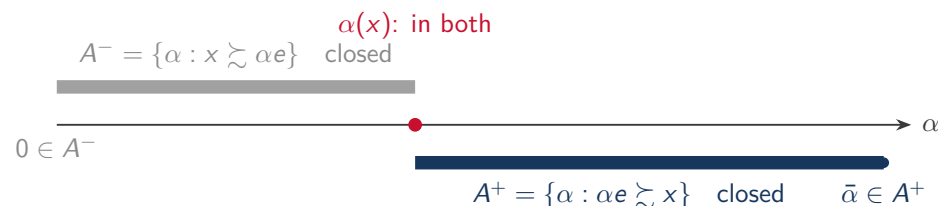
Why $\alpha(x)$ exists and is unique. Fix x and split the ray:

$A^- = \{\alpha \geq 0 : x \succsim \alpha e\}$ and $A^+ = \{\alpha \geq 0 : \alpha e \succsim x\}$. Both are closed, by continuity (slide 13). Both are nonempty: $0 \in A^-$ since $x \succsim 0$, and any $\bar{\alpha}$ with $\bar{\alpha}e \gg x$ is in A^+ , by monotonicity. By completeness they cover $\mathbb{R}_+$. Two nonempty closed sets covering the connected set $\mathbb{R}_+$ must overlap, so some α has $\alpha e \sim x$.

Uniqueness: if $\alpha' > \alpha$ then $\alpha'e \gg \alpha e$, so $\alpha'e \succ \alpha e$ by monotonicity; at most one point of the ray is indifferent to x . And u represents $\succsim$: $x \succsim y$ exactly when $\alpha(x)e \succsim \alpha(y)e$, exactly when $\alpha(x) \geq \alpha(y)$.

Where each assumption enters: continuity closes the two sets; monotonicity fills them and orders the ray. That u is itself continuous is MWG pp. 48–49, skipped.

Why every curve crosses the diagonal, once



closed (continuous), nonempty (monotone), cover $\mathbb{R}_+$ (complete) $\Rightarrow$ they overlap

monotone: $\alpha' > \alpha \Rightarrow \alpha'e \succ \alpha e$, so they overlap in **one** point

$$u(x) \geq u(y) \iff \alpha(x)e \succsim \alpha(y)e \iff x \succsim y$$

MWG, proof of Prop. 3.C.1, pp. 47–48. $\mathbb{R}_+$ is connected: two nonempty closed sets covering it meet.

In pairs: continuous? represented by some u ?

$x \succsim y \iff$	continuous	some u	
$x_1 x_2 \geq y_1 y_2$	✓	✓	$u = x_1 x_2$
$x_1 > y_1$, or $x_1 = y_1$ and $x_2 \geq y_2$	✗	✗	slides 14–15
$\lfloor x_1 \rfloor \geq \lfloor y_1 \rfloor$ (whole units of good 1 only)	✗	✓	$u = \lfloor x_1 \rfloor$
$\ x - (5, 5)\ \leq \ y - (5, 5)\ $	✓	✓	$u = -\ x - (5, 5)\ $

Six minutes. For “not continuous”, give a sequence that reverses at the limit.

Row 3: $(2 - \frac{1}{n}, 0) \succsim (1, 0)$ for all n , the limit $(2, 0)$ is not; continuity is **sufficient**, not necessary.

Row 4: closed discs; not monotone, and it does not matter.

Slide 18 In pairs: continuous? represented by some u ?

Try it: for each preference, is it continuous, and is it represented by some u ? For “not continuous”, give a sequence that reverses at the limit.

Row 1, $x_1 x_2 \geq y_1 y_2$: $u = x_1 x_2$ represents it, and u is continuous, so limits pass through u and the preference is continuous. Row 2, lexicographic: neither, slides 14–15.

Row 3, only whole units of good 1 count: $(2 - \frac{1}{n}, 0)$ is no better than $(1, 0)$ for every n (both floors are 1), but the limit $(2, 0)$ has floor 2 and is strictly better: not continuous. Yet $u = \lfloor x_1 \rfloor$ represents it. Lesson: continuity is sufficient for a utility function, not necessary.

Row 4, closer to $(5, 5)$ is better: $u = -\|x - (5, 5)\|$ represents it and is continuous, so the preference is continuous (the upper contour set of x is a closed disc round $(5, 5)$; the lower contour set is everything on or outside that circle, also closed). It is not monotone, so the diagonal construction of slide 16 fails here, but the general theorem (Debreu, 1954) does not need monotonicity.

Slide 19 Break

Ten minutes.

Break

Back at 11:00

Quasiconcavity and concavity

MWG 3.C

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Slide 20 Quasiconcavity and concavity

Part 3: what the shape of u says about $\succsim$, and what it does not.
MWG 3.C.

Definition: quasiconcave utility

u is **quasiconcave** if every set
 $\{y \in X : u(y) \geq u(x)\}$ is convex, that is,

$$u(\alpha x + (1 - \alpha)y) \geq \min\{u(x), u(y)\}$$

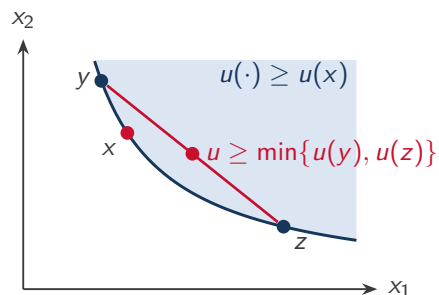
for all $x, y \in X$ and $\alpha \in [0, 1]$; **strictly**: $>$ when
 $x \neq y$, $\alpha \in (0, 1)$

$\succsim$ convex $\iff u$ quasiconcave

strictly convex $\iff$ strictly quasiconcave

MWG 3.C, pp. 49–50.

Class 2's picture: the upper contour sets of $\succsim$ are the sets $\{u \geq u(x)\}$.



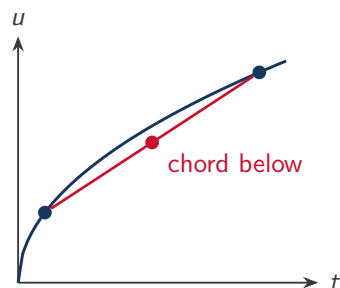
Slide 21 Definition: quasiconcave utility

MWG 3.C, pp. 49–50. u is quasiconcave if every set $\{y : u(y) \geq u(x)\}$ is convex; equivalently, the utility of a mix $\alpha x + (1 - \alpha)y$ is at least the smaller of $u(x)$ and $u(y)$. Strictly quasiconcave: strictly larger, for $x \neq y$ and $0 < \alpha < 1$.

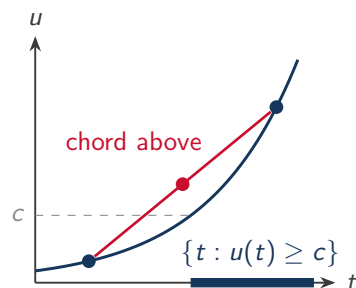
Those sets are exactly the upper contour sets of $\succsim$ (class 2's picture, now with u -values). So $\succsim$ is convex if and only if u is quasiconcave, and strictly convex if and only if u is strictly quasiconcave.

Is “ u is quasiconcave” a statement about u or about $\succsim$? About $\succsim$: it holds for every u that represents $\succsim$, or for none. Nothing new about the preference, just a name for the function.

Concave is stronger, and it is about the numbers



concave: $u(t) = \sqrt{t}$
and quasiconcave



quasiconcave, not concave: $u(t) = e^{0.6t}$
increasing, so every level set is a half-line: convex

concave: $u(\alpha x + (1 - \alpha)y) \geq \alpha u(x) + (1 - \alpha)u(y) \Rightarrow$ quasiconcave, not $\Leftarrow$
MWG 3.C, p. 50: some convex $\succsim$ have no concave representation at all.

Slide 22 Concave is stronger, and it is about the numbers

u is concave if the utility of a mix is at least the mix of the utilities: $u(\alpha x + (1 - \alpha)y) \geq \alpha u(x) + (1 - \alpha)u(y)$. Concave implies quasiconcave, because an average of two numbers is at least their minimum. The converse fails.

Left: $u(t) = \sqrt{t}$, concave, the chord lies below the graph. Right: $u(t) = e^{0.6t}$; every level set $\{t : u(t) \geq c\}$ is a half-line, convex, so u is quasiconcave, but the chord lies above the graph: not concave.

Concavity compares utility numbers, the score of the average against the average of the scores: cardinal. Quasiconcavity looks only at the level sets: ordinal. MWG p. 50: a convex $\succsim$ need not have a concave u , and some convex preferences have no concave representation at all.

Calculus check for a twice-differentiable u : concave if and only if the Hessian is negative semidefinite everywhere; in two variables, $u_{11} \leq 0$, $u_{22} \leq 0$ and $u_{11}u_{22} - u_{12}^2 \geq 0$.

Slide 23 What survives a strictly increasing relabelling $v = f \circ u$

$v = f \circ u$

Replace u by $v = f \circ u$ with f strictly increasing. The top rows survive: v represents the same $\succsim$ (that is what strictly increasing means); v is increasing in the goods when $\succsim$ is monotone; v is quasiconcave when u is, since the level sets are the same; and the MRS at every bundle is unchanged, because $v_1/v_2 = f'(u)u_1/f'(u)u_2 = u_1/u_2$. The slope of the curve is a fact about the map.

The bottom rows do not survive: concavity (next slide, x_1x_2 against $\sqrt{x_1x_2}$); marginal utility, rescaled by $f'(u)$; even continuity, since a strictly increasing f with a jump, say $f(t) = t$ below 4 and $t + 1$ from 4 on, makes v jump (MWG footnote 6).

Top: ordinal, facts about $\succsim$. Bottom: cardinal, properties of one convenient representation.

What survives a strictly increasing relabelling $v = f \circ u$

	survives	
represents the same $\succsim$	✓	that is what f increasing means
increasing ($x \gg y \Rightarrow v(x) > v(y)$)	✓	monotone $\succsim$
quasiconcave	✓	same level sets, same $\succsim$: convex $\succsim$
the MRS at every bundle	✓	$f'(u)u_1/f'(u)u_2 = u_1/u_2$; f differentiable, $f' > 0$
concave	✗	x_1x_2 vs $\sqrt{x_1x_2}$: next slide
marginal utility $\partial u/\partial x_1$	✗	rescaled by f'
continuous	✗	a jump in f makes v jump

Top: **ordinal**, facts about $\succsim$. Bottom: **cardinal**, facts about one convenient u .

MWG 3.C, p. 50 and footnote 6.

In pairs: quasiconcave? concave?

$u(x) =$	quasiconcave	concave	
$x_1 x_2$	✓	✗	$u(1, 1) = 1 < 2 = \frac{1}{2}u(0, 0) + \frac{1}{2}u(2, 2)$
$\sqrt{x_1 x_2}$	✓	✓	same $\succsim$ as row 1; the geometric mean is concave
$x_1 + x_2$	✓	✓	linear: both, neither strictly
$(x_1 + x_2)^2$	✓	✗	same $\succsim$ as row 3; $u(1, 1) = 4 < 8$

Five minutes. Test concavity at $(0, 0)$, $(2, 2)$ and their midpoint.

Rows 1–2 and 3–4 are the same preferences: quasiconcavity agrees, concavity does not.

Slide 24 In pairs: quasiconcave? concave?

Try it: is each u quasiconcave, and is it concave? Test concavity at $(0, 0)$, $(2, 2)$ and their midpoint $(1, 1)$.

$x_1 x_2$: quasiconcave (Cobb–Douglas type, a convex preference from class 2), not concave: $u(1, 1) = 1$, below the average $\frac{1}{2} \cdot 0 + \frac{1}{2} \cdot 4 = 2$.

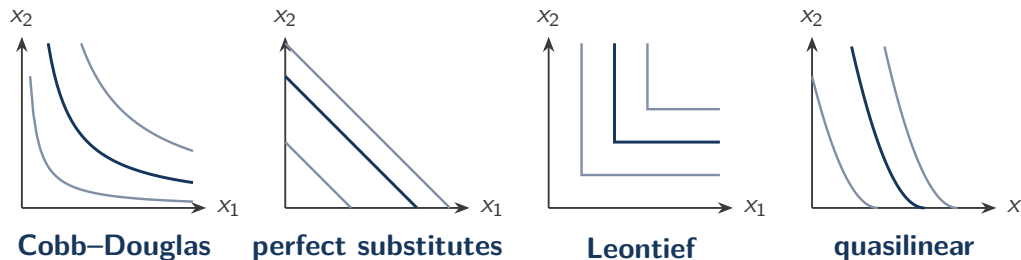
$\sqrt{x_1 x_2}$: quasiconcave, the same preference; and concave, because the geometric mean is concave (its Hessian is negative semidefinite: $u_{11} < 0$, determinant 0). The midpoint test gives $1 \geq 1$, consistent but not a proof.

$x_1 + x_2$: linear, so both concave and quasiconcave, neither strictly.

$(x_1 + x_2)^2$: quasiconcave (an increasing transformation of a linear function; its level sets are half-planes) but not concave: $u(1, 1) = 4$, below $\frac{1}{2} \cdot 0 + \frac{1}{2} \cdot 16 = 8$.

Rows 1–2 and rows 3–4 are the same preferences: quasiconcavity agrees within each pair, concavity does not. That is slide 23 in one table.

Four utility functions you will use all month



$\alpha, \beta, a, b > 0$; φ increasing and concave. $MRS_{12} = u_1/u_2$, class 2. Leontief kinks: MWG Figure 3.C.3.

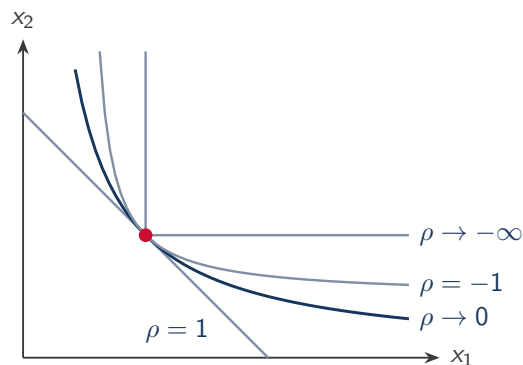
Slide 25 Four utility functions you will use all month

Cobb–Douglas $x_1^\alpha x_2^\beta$: $MRS = \alpha x_2 / (\beta x_1)$, falling along a curve as x_1 rises. Perfect substitutes $ax_1 + bx_2$: straight lines, $MRS = a/b$, constant.

Leontief $\min\{ax_1, bx_2\}$: L-shaped curves with the kink on the line $x_2 = (a/b)x_1$; no MRS at the kink, since u is not differentiable there (MWG Figure 3.C.3: a continuous preference need not have a differentiable u). Quasilinear $x_1 + \varphi(x_2)$ with φ increasing and concave: $u_1 = 1$ and $u_2 = \varphi'(x_2)$, so $MRS = 1/\varphi'(x_2)$ depends on x_2 only, and the curves are horizontal translates of each other, class 2's picture.

Check each against the properties you know: monotone, strongly monotone, convex, strictly convex.

One formula holds the whole family: CES



$$u(x) = \left(\frac{1}{2}x_1^\rho + \frac{1}{2}x_2^\rho \right)^{1/\rho}$$

$\rho \leq 1, \rho \neq 0$; for $\rho < 0$ take $x \gg 0$

$\rho = 1$: perfect substitutes $\frac{1}{2}(x_1 + x_2)$

$\rho \rightarrow 0$: Cobb–Douglas $\sqrt{x_1 x_2}$

$\rho \rightarrow -\infty$: Leontief $\min\{x_1, x_2\}$

All four curves are $u = 2$. Lower ρ : a sharper bend, the goods substitute less easily.

Elasticity of substitution $\sigma = 1/(1 - \rho)$; Muñoz-García §1.8.4 (with weights a, b). Returns with demand.

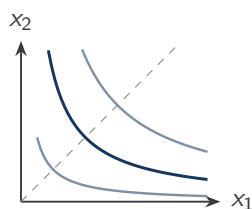
Slide 26 One formula holds the whole family: CES

One formula holds the whole family: $u(x) = (\frac{1}{2}x_1^\rho + \frac{1}{2}x_2^\rho)^{1/\rho}$ with $\rho \leq 1, \rho \neq 0$; for $\rho < 0$ take $x \gg 0$, since the formula is undefined on the axes.

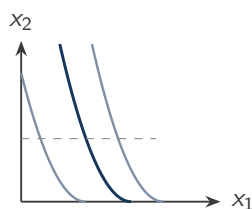
$\rho = 1$ gives $\frac{1}{2}(x_1 + x_2)$, perfect substitutes. As $\rho \rightarrow 0$ the function tends to $\sqrt{x_1 x_2}$, Cobb–Douglas (take logs and use l'Hôpital). As $\rho \rightarrow -\infty$ it tends to $\min\{x_1, x_2\}$, Leontief. Without the weights $\frac{1}{2}$ the function itself blows up or vanishes as $\rho \rightarrow 0$ and only the ranking has a limit: the ordinal point again.

With these weights $u(2, 2) = 2$ for every ρ , so all four curves drawn are $u = 2$: the line $x_1 + x_2 = 4$, the hyperbola $x_1 x_2 = 4$, for $\rho = -1$ the curve $1/x_1 + 1/x_2 = 1$, and the L. Lower ρ , sharper bend: the goods substitute less easily. The elasticity of substitution is $\sigma = 1/(1 - \rho)$: $\infty, 1$ and 0 at the three special cases. Returns with demand.

Class 2's two families, in utility form



homothetic



quasilinear in good 1

continuous $\succsim$: **homothetic** $\iff$ some u with $u(\alpha x) = \alpha u(x)$ for $\alpha > 0$
 $\sqrt{x_1 x_2}, x_1 + x_2, \min\{x_1, x_2\}$

continuous $\succsim$: **quasilinear** in good 1 $\iff$ some u of the form $x_1 + \varphi(x_2)$
 $x_1 + \sqrt{x_2}, x_1 + \ln(1 + x_2)$

“Some u ”: a cardinal choice. $x_1 x_2$ is homothetic too, and not homogeneous of degree one.

MWG Exercise 3.C.5, stated on p. 50; quasilinear on $X = \mathbb{R} \times \mathbb{R}_+$. Both return with demand next week.

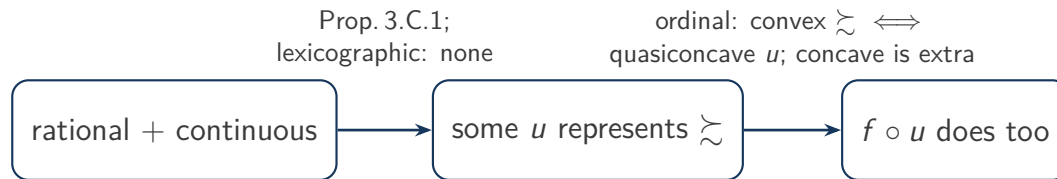
Slide 27 Class 2's two families, in utility form

Class 2's two families in utility form (MWG Exercise 3.C.5). A continuous $\succsim$ on $\mathbb{R}_+^2$ is homothetic if and only if it has some utility function homogeneous of degree one, $u(\alpha x) = \alpha u(x)$ for $\alpha > 0$: $\sqrt{x_1 x_2}, x_1 + x_2, \min\{x_1, x_2\}$.

A continuous $\succsim$ on $\mathbb{R} \times \mathbb{R}_+$ is quasilinear in good 1 if and only if it has some utility function of the form $x_1 + \varphi(x_2)$: $x_1 + \sqrt{x_2}, x_1 + \ln(1 + x_2)$.

“Some u ” is a cardinal choice, not every representation: $x_1 x_2$ is homothetic too (the same preference as $\sqrt{x_1 x_2}$) but homogeneous of degree two. “Continuous” is there for a reason: the lexicographic preference satisfies both definitions from class 2 and has no utility function at all (slide 15). Both families return with demand next week.

Today in one line



Wednesday: budget set, utility maximisation, Walrasian demand.

Problem set 1: one copy per group, due Wed 14 Oct.

Read MWG 2.D and 3.D (or Muñoz-Garcia ch. 2) for Wednesday 7 October.

Slide 28 Today in one line

Rational and continuous gives some u that represents $\succsim$ (Proposition 3.C.1); the lexicographic preference shows both that the hypothesis can fail and that some preferences have no u at all. Any strictly increasing $f \circ u$ is as good as u : utility is ordinal. Convex $\succsim$ means quasiconcave u ; concavity is an extra, cardinal property of one representation.

Wednesday 7 October: the budget set, utility maximisation and Walrasian demand. Reading: MWG 2.D and 3.D, or Muñoz-Garcia chapter 2.

Problem set 1 goes out now, one copy per group, due in class on Wednesday 14 October.